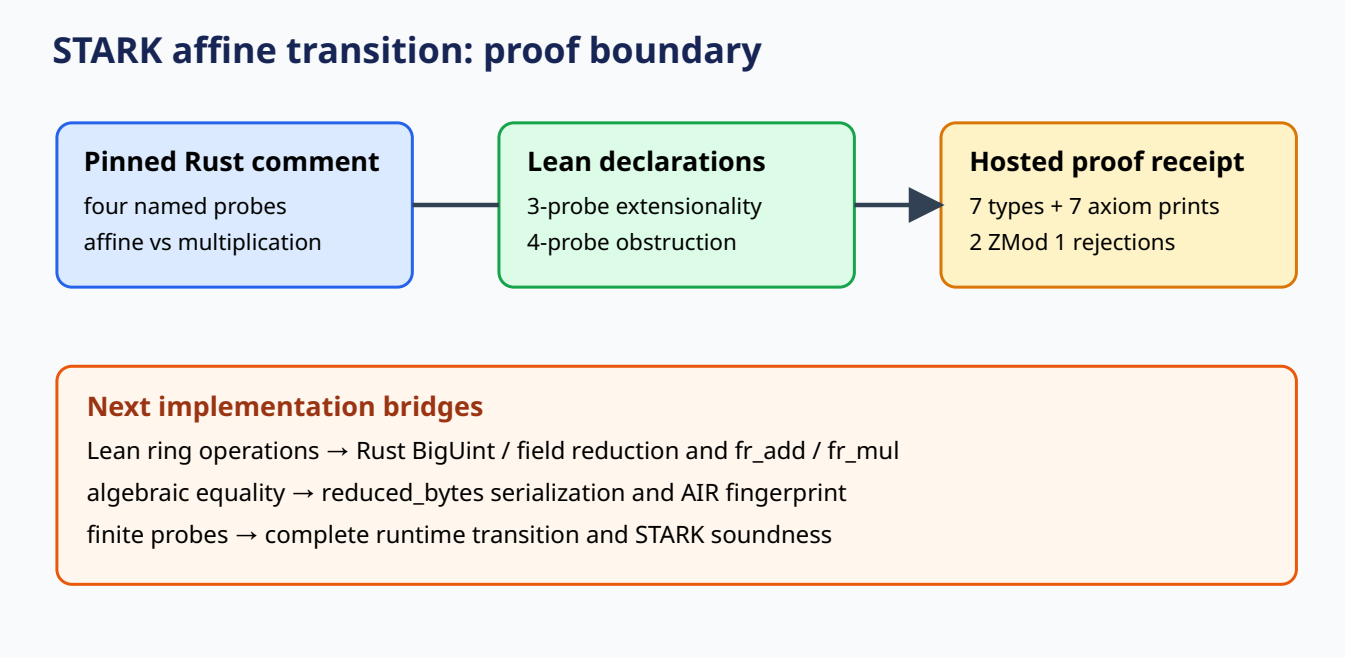
STARK affine transition proof review

2026-11-03

Scoped result. The seventh Lean theorem proves the pinned Rust comment's four-point obstruction for affine ring expressions over every nontrivial commutative ring. The original six declarations are unchanged. The three-probe theorem establishes coefficient extensionality.

Independent evidence. Prior sealed J1 blind derivation holds. J2 checked current source and elaborated types. J3 independently downloaded hosted artifact 13264657224 and passed 23 receipt consistency checks: seven types and axiom prints, two rejected zero-ring applications, 89 workspace inputs, ten verifier inputs, and ten stable dependency pins. The named GitHub Lean job succeeded.

Verdict. Machine status: passed on the named hosted workflow. Fidelity: faithful for the stated affine-expression claims. Review outcome: validated at that scope. Receipt consistency alone is not an independent binary or kernel recheck. The expression theorem does not establish Rust reduction, byte encoding, complete AIR behavior, or STARK soundness.



Source and audit details: REVIEW.md and the three frozen reviewer reports in this packet. The diagram shows mathematical and receipt correspondence above, with implementation bridges below.